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Collection name

patent_rag_chunks

Patent results

20

Candidate chunks

100

Use cross-encoder reranking

Reranker scores are raw logits, not probabilities.

Patent Intelligence

Inspectable patent search over the existing ChromaDB index

Describe the invention or feature you are looking for

Coating

Search patents

Results

Query: Coating

Collection chunks	Candidates	Patents shown	Embedding	Vector search
67,303	100	20	0.14s	0.59s

Cross-encoder reranking: 2.92s

Retrieved chunks	Unique patents	Returned patents	Reranked
100	26	20	Yes

rank	patent_number	title	score	evidence_chunks	max_similarity	top3_mean_similarity	max_rer
11	14903825	METHOD FOR PREPARING A COATING OF A CYLINDRICAL SURFACE OF A SCANNING SL	2.1232	3	0.0207	-0.0288	
12	14989677	STABLE BENZIMIDAZOLE FORMULATION	2.0621	2	-0.0321	-0.049	
13	14991773	COMPOSITIONS COMPRISING DEHYDRATED MICRO-ORGANISMS, METHOD OF PREPA	1.8023	1	-0.0172	-0.0172	
14	14992348	POLARIZING PLATE AND METHOD FOR MANUFACTURING POLARIZING PLATE	1.1176	3	0.0308	0.0137	
15	15002388	NOVEL PULSATILE DRUG DELIVERY SYSTEM	0.995	1	-0.0553	-0.0553	
16	14994969	Polymer Composite Vacuum Components	0.8934	1	-0.0307	-0.0307	
17	14904359	Nicotine Salt with Meta-Salicylic Acid	0.2273	1	0.0025	0.0025	
18	14989141	POLLUTION CONTROL DEVICE STRUCTURE WITH LOWER FRICTION SURFACE AND UN	0.1273	2	-0.0074	-0.0363	
19	14903888	PLATED POLYMERS WITH INTUMESCENT COMPOSITIONS AND TEMPERATURE INDICA	0.0406	2	-0.0053	-0.0321	
20	14988948	THIN FILM DEPOSITION	-0.3458	1	-0.0131	-0.0131	

> 1. 14905300 — Composite Coatings of Oxidized and/or Phosphorous Copper

> 2. 14903677 — Method for Coating Metal Surfaces of Substrates and Objects Coated According to Said Method

> 3. 14991299 — TWO-COMPONENT DEVELOPER FOR ELECTROSTATIC LATENT IMAGE DEVELOPMENT

> 4. 14906540 — PHOTO-ACTIVATED HYDROPHILIC COATINGS AND METHODS FOR THEIR PREPARATION AND USE

▼ 5. 14906106 — PRESSURE-SENSITIVE ADHESIVE TAPES FOR THE ADHESIVE BONDING OF WINDOWS MORE PARTICULARLY IN MOBILE EQUIPMENT

Patent score	Max similarity	Top-3 mean	Evidence chunks	Decision
3.6696	-0.0464	-0.0464	1	PENDING
Max reranker score	Top-3 mean reranker score			
3.6696	3.6696			

Retrieved 1 chunk(s) for this patent; showing the top 3 evidence chunks.

Chunk 71 · embedding rank 82 · similarity -0.0464 · reranker rank 11 · score 3.6696

in the case of adhesive tapes with a permanent backing, coating can take place directly onto the backing, or the adhesive is likewise first coated onto a temporary backing and then transferred from there to the permanent backing by lamination. suitable as permanent backings are the conventional materials, such as, for example, plastics films, particularly preferably films of transparent, translucent, colored, white or black plastics material such as in particular polyester, especially polyethylene terephthalate. backing thicknesses are typically from 1 μm to 250 μm, preferably from 10 μm to 150 μm. before the coating, the solvent (mixture) may optionally be removed. coating then takes place in a solvent - free manner. coating can also take place in a solvent - containing manner. the formulation is then dried after the coating. the solvent - containing formulation is advantageously stirred or otherwise moved in convection before being fed to the coating unit, very advantageously immediately before being fed to the coating unit. the formulation can also be produced in a solvent - free manner. to that end, a thread of solvent - free polyacrylate, for example, is fed to a

▼ 6. 15002057 — POSITIVE ACTIVE MATERIAL AND LITHIUM SECONDARY BATTERY INCLUDING POSITIVE ELECTRODE THAT INCLUDES THE POSITIVE ACTIVE MATERIAL

Patent score	Max similarity	Top-3 mean	Evidence chunks	Decision
3.4946	0.0273	0.0004	3	ACCEPTED
Max reranker score			Top-3 mean reranker score	
4.2834			2.7058	
Retrieved 3 chunk(s) for this patent; showing the top 3 evidence chunks.				
Chunk 13 · embedding rank 32 · similarity 0.0273 · reranker rank 9 · score 4.2834				
mnf3, ndf3, vof3, prf3, sbf3, scf3, smf3, tbf3, tif3, tmf3, yf3, ybf3, tif3, cef4, gef4, hff4, sif4, snf4, tif4, vf4, zrf4, nbf5, sbf5, taf5, bif5, mof6, ref6, sf6, and wf6. in some embodiments, the coating layer may include compound of a coating element selected from the group consisting of oxide, hydroxide, oxyhydroxide, oxycarbonate, and hydroxycarbonate of the coating element. these compounds for the coating layer may be amorphous or crystalline. the coating element for the coating layer may be scandium (sc), yttrium (y), niobium (nb), chromium (cr), molybdenum (mo), tungsten (w), manganese (mn), iron (				
Chunk 15 · embedding rank 54 · similarity -0.0112 · reranker rank 39 · score 1.9703				
active material when a compound of the coating element is used. for example, the coating layer may be formed using a spray coating method, a dipping method, or the like. in some embodiments, the coating layer may be a continuous layer or a discontinuous layer, for example, in the form of islands. hereinafter, a method of preparing a positive active material of formula 1 according to an embodiment will be described. methods for preparing the positive active material of formula 1 are not limited, may include, for example, a co - precipitation method, a solid - phase method, and the like. first, the positive active material of formula 1 may be prepared using a co - precipitation method as follows. a positive active material precursor represented by formula 2 as a positive active material precursor may be mixed with a lithium precursor and an alkali metal source, and then thermally treated at a temperature of about 400° c. to about 1200° c. in the air or oxygen atmosphere to obtain the positive active material represented by formula 1. nixcoymn1 - x - y (oh) 2 formula 2 wherein, in formula 2, a may be				
Chunk 14 · embedding rank 56 · similarity -0.0149 · reranker rank 42 · score 1.8638				
), niobium (nb), chromium (cr), molybdenum (mo), tungsten (w), manganese (mn), iron (fe), boron (b), indium (in), carbon (c), stibium (sb), lanthanum (la), cerium (ce), samarium (sm), gadolinium (gd), magnesium (mg), aluminum (al), cobalt (co), potassium (k), sodium (na), calcium (ca), silicon (si), titanium (ti), vanadium (v), tin (sn), germanium (ge), gallium (ga), arsenic (as), zirconium (zr), or mixtures thereof. the coating layer may be formed using any method that does not adversely affect the physical properties of the positive active material when a compound of the coating element is used. for example, the coating layer may be formed using a spray coating method, a dipping method, or the like. in some embodiment				

▼ 7. 14906379 — FABRIC PRINT MEDIUM

Patent score	Max similarity	Top-3 mean	Evidence chunks	Decision
3.3455	0.0205	-0.0100	3	PENDING
Max reranker score			Top-3 mean reranker score	
3.6857			3.0052	

Retrieved 3 chunk(s) for this patent; showing the top 3 evidence chunks.

Chunk 42 · embedding rank 95 · similarity -0.0641 · reranker rank 10 · score 3.6857

form a coating layer, said composition including a non - halogenated flame retardant agent having phosphorous - containing ingredient and nitrogen - containing ingredient compounded at a 1 : 1 ratio, a water - soluble polymer binder and a water - soluble high - valence metal complex ; and drying the fabric substrate under heat to form a fabric print medium. the coating compositions can be prepared in a liquid carrier in order to disperse or solubilize coating composition components. such liquid carrier is, for example, an aqueous solvent such as water and low boiling point alcohol. the liquid carrier can be removed, at least in part, from the final product once the coating composition is applied to the fabric. the liquid carrier may include water, co - solvents, surfactants, viscosity modifying agents, inorganic ingredients, ph control agents and / or deformers. the primary function of the carrier is to dissolve / disperse and / or carry the solids or other components that remain on the fabric as a coating, and to provide a carrier that will suitably carry all the components in the composition and help them uniformly distribute on the fabric base surface. there is no specific

Chunk 6 · embedding rank 38 · similarity 0.0137 · reranker rank 25 · score 2.7011

. the present disclosure refers to a fabric print medium comprising a fabric base substrate and a coating composition applied to said fabric base substrate. the coating composition includes a non - halogenated flame retardant agent having phosphorous - containing ingredient and nitrogen - containing ingredient compounded at a 1 : 1 ratio ; a water - soluble polymer binder and a water - soluble high - valence metal complex. without being linked by any theory, it is believed that the coating composition, also called treatment composition, once applied on the fabric base substrate, forms a thin layer onto the fabric base surface. said thin layer has a first structure before image formation on the fabric (e. g., using inkjet printing for example) and a second different structure once the ink has been applied. the first structure is configured to allow ink colorants to adhere better onto the structure. the first structure formed from the treatment composition is transformed into a second structure, during printing process, to further protect the image after image formation. in some examples, the fabric base substrate has two sides, and both of the two sides are coated with the coating composition. fig. 1 and fig. 2 illustrate the fabric print medium

Chunk 7 · embedding rank 34 · similarity 0.0205 · reranker rank 26 · score 2.6288

image formation. in some examples, the fabric base substrate has two sides, and both of the two sides are coated with the coating composition. fig. 1 and fig. 2 illustrate the fabric print medium (100) as described herein. as illustrated in fig. 1, the fabric print medium (100) encompasses a fabric base substrate (110) and a coating composition or layer (120). the coating composition (120) is applied on one side of the bottom supporting substrate (110). if said coated side is used as

an image - receiving side, the other side, i. e. backside, may not have any coating at all, or may be coated with other chemicals (e. g. sizing agents and backing adhesives) or coatings, or laminate with other materials such as backing paper and plastic film / sheet to meet certain features such as to balance the curl of the final product or to improve sheet feeding in printer. in some other examples, such as illustrated in fig. 2, the coating composition (120) is applied to both opposing sides of the supporting fabric base substrate (110). the double - side coated media has thus a sandwich structure,

8. 14902959 — OPTICAL FILM

Patent score	Max similarity	Top-3 mean	Evidence chunks	Decision
3.1317	0.0741	0.0741	1	PENDING
Max reranker score			Top-3 mean reranker score	
3.1317			3.1317	

Retrieved 1 chunk(s) for this patent; showing the top 3 evidence chunks.

Chunk 26 · embedding rank 20 · similarity 0.0741 · reranker rank 18 · score 3.1317

coating composition may include various dyes and pigments in order to absorb ultraviolet rays, infrared rays, or visible rays, as required. in certain cases, a viscosity modifier, such as a thickener or filler, may be properly added. the tn composition may be applied by all kinds of liquid - coating methods. in some embodimENTS, after coating, the tn composition is polymerized, or converted to a tn layer. the conversion may be achieved by evaporating the solvent and heating to align the tn material ; crosslinking the tn composition ; or, for example, applying heat such as actinic irradiation ; radiating light, such as ultraviolet rays, visible rays, or infrared rays, and radiating electron beams, or combinations thereof, or various techniques including curing of the tn composition by use of similar techniques. in the embodiment, the tn composition may include the compound of formula 1, a polymerization initiator, and a chiral agent. as the polymerization initiator, for example, a radical initiator generating radicals by heat or light may be used. the radical initiator initiates polymerization or cross

9. 15001258 — WATER-BASED ANTI-SOILING AGENT, ANTI-SOILING LAYER, LAYERED BODY, AND SOLAR BATTERY MODULE

Patent score	Max similarity	Top-3 mean	Evidence chunks	Decision
2.4500	0.0960	0.0520	5	PENDING
Max reranker score			Top-3 mean reranker score	
3.0924			1.8077	

Retrieved 5 chunk(s) for this patent; showing the top 3 evidence chunks.

Chunk 39 · embedding rank 17 · similarity 0.0960 · reranker rank 20 · score 3.0924

properties. in this manner, the anti - soiling coating layer according to the invention is preferably provided, for example, on the glass surface on the light receiving surface side of the solar battery module, and is useful as an anti - soiling coating layer for a solar battery module. the method of coating the water - based anti - soiling agent according to the invention is not particularly limited, and examples thereof include spray coating, brush coating, roller coating, bar coating, and dip coating. in the drying method after coating, drying may be performed at room temperature, or heating may be performed at 40° c. to 120° c. for about 1 minute to 30 minutes. (physical properties of anti - soiling coating layer) in order to cause the anti - soiling coating layer according to the invention to exhibit sufficient anti - soiling properties, it is preferable that the surface of the anti - soiling coating layer exhibits antistatic properties and hydrophilicity. accordingly, it is possible to suppress the attachment of the contaminants, and also even if contaminants are attached to the surface of the anti - soiling coating layer, the stain has to

Chunk 38 · embedding rank 80 · similarity -0.0457 · reranker rank 31 · score 2.4015

component. accordingly, the silica fine particles are preferably added in the latter half of the manufacturing of the liquid, or at the end. at this point, if the dispersion liquid of the silica fine particles is used, both ph of the dispersion liquid and ph of the coating agent may be acidic or basic. (forming method of anti - soiling coating layer) the anti - soiling coating layer according to the invention can be formed by coating and drying the water - based anti - soiling agent described above. the target of coating the water - based anti - soiling agent is not particularly limited, and suitably used on surfaces of various supports such as glass, a resin, metal, and ceramics. in addition, if a glass layer is used as a base material, the condensation of the hydroxyl group on silicon is generated between hydroxyl groups of the glass surface, and thus it is possible to obtain a layered body having excellent adhesive properties. in this manner, the anti - soiling coating layer according to the invention is preferably provided, for example, on the glass surface on the light receiving surface side of the solar battery module

Chunk 34 · embedding rank 76 · similarity -0.0415 · reranker rank 70 · score -0.0708

of a water - based anti - soiling agent, and is used as a catalyst for the condensation of the siloxane oligomer. (other additives) the water - based anti - soiling agent according to the invention may appropriately have an additive such as preservative, in addition to the above. meanwhile, the water - based anti - soiling agent according to the invention does not require light irradiation or a high - temperature heat treatment at the time of forming a film after coating, and thus does not require a photopolymerization initiator or a themo polymerization initiator corresponding thereto. instead, considering the storage stability of the coating agent, it is rather preferable not to have a photopolymerization initiator or a theimo polymerization initiator. in this manner, according to the invention, it is possible to form an anti - soiling coating layer from a water - based anti - soiling agent in a simple method. in addition, most of the solid content of the anti - soiling agent according to the invention is silicon and oxygen, and the carbon content is small. if the carbon

10. 14994425 — PROCESS CHAMBER FOR CYCLIC AND SELECTIVE MATERIAL REMOVAL AND ETCHING

11. 14903825 — METHOD FOR PREPARING A COATING OF A CYLINDRICAL SURFACE OF A SCANNING SLEEVE FOR A PRINTING MACHINE AND SCANNING SLEEVE THAT IS PREPARED BY THIS METHOD

> 12. 14989677 — STABLE BENZIMIDAZOLE FORMULATION

> 13. 14991773 — COMPOSITIONS COMPRISING DEHYDRATED MICRO-ORGANISMS, METHOD OF PREPARATION THEREOF, AND USES THEREOF

> 14. 14992348 — POLARIZING PLATE AND METHOD FOR MANUFACTURING POLARIZING PLATE

> 15. 15002388 — NOVEL PULSATILE DRUG DELIVERY SYSTEM

> 16. 14994969 — Polymer Composite Vacuum Components

> 17. 14904359 — Nicotine Salt with Meta-Salicylic Acid

> 18. 14989141 — POLLUTION CONTROL DEVICE STRUCTURE WITH LOWER FRICTION SURFACE AND UNDERLYING HIGHER FRICTION SURFACE

▼ 19. 14903888 — PLATED POLYMERS WITH INTUMESCENT COMPOSITIONS AND TEMPERATURE INDICATORS

Patent score	Max similarity	Top-3 mean	Evidence chunks	Decision
0.0406	-0.0053	-0.0321	2	PENDING
Max reranker score	Top-3 mean reranker score			
0.7973	-0.7162			

Retrieved 2 chunk(s) for this patent; showing the top 3 evidence chunks.

Chunk 21 · embedding rank 50 · similarity -0.0053 · reranker rank 58 · score 0.7973

244 (or the glass - transition temperature of the polymer substrate 244), a suitable temperature - indicating coating 242 may be selected according to a block 254, as shown. according to a next block 256, the selected temperature - indicating coating may be applied to selected surfaces of the metal plating 246 and / or outer surfaces of the polymer substrate 244. in this regard, the selected surfaces may be surfaces of the component which have a greater probability for exposure to high temperatures during in - service operation. the temperature - indicating coating 242 may be applied as a film - coating, as a tape, or as a spray coating which is brushed on or painted on, although other application methods may also be used. industrial applicability from the foregoing, it can therefore be seen that the present disclosure can find industrial applicability in many situations, including, but not limited to, situations utilizing metal - plated or metal - coated polymer materials which may be exposed to high temperatures or temperatures above their design limits. as disclosed herein, the polymer support structure may be formed from a lightweight polymer and the metal plating (s) applied to its surfaces may substantially contribute to the structural resilience of the

Chunk 20 · embedding rank 90 · similarity -0.0588 · reranker rank 82 · score -2.2296

(liquid bed, powder bed, deposition processes), or composite layup (autoclave, compression, or liquid molding). one or more metal plating layers 246 may then be applied to selected outer surfaces of the polymer substrate 244 according to a next block 252. the block 252 may involve first suitably activating and metallizing the selected outer surfaces of the polymer substrate 244 using techniques that are well - established in the industry. once the selected outer surfaces have been suitably activated and metallized, the metal plating layers may be deposited using metal deposition methods apparent to those having ordinary skill in the art such as electrolytic plating, electroless plating, electroforming, or another suitable deposition method. alternatively, the metal plating 246 may be applied to selected outer surfaces of the polymer substrate 244 as a coating by spraying or other deposition processes. based on the known critical temperature or set of critical temperatures of the bond between the metal plating 246 and the polymer substrate 244 (or the glass - transition temperature of the polymer substrate 244), a suitable temperature - indicating coating 242 may be selected according to a block 254, as shown. according to a next block 256

▼ 20. 14988948 — THIN FILM DEPOSITION

Patent score	Max similarity	Top-3 mean	Evidence chunks	Decision
-0.3458	-0.0131	-0.0131	1	ACCEPTED
Max reranker score	Top-3 mean reranker score			
-0.3458	-0.3458			

Retrieved 1 chunk(s) for this patent; showing the top 3 evidence chunks.

Chunk 14 · embedding rank 55 · similarity -0.0131 · reranker rank 74 · score -0.3458

and greater than an affinity of the surface for the liquid. the material / liquid may, for example, include acrylate compounds or epoxy compounds. the material may, for example, be applied by inkjet printing or drop coating to protrusions for which a measurable aspect exceeds a threshold value. in a number of embodiments wherein the surface of the protrusion adjacent to the shadowed region is caused to have affinity for the liquid greater than an affinity of the remainder of the surface of the protrusion for the liquid and greater than an affinity of the surface for the liquid, the method further includes applying the liquid to at least a portion of the surface including a plurality of protrusions and removing a bulk of the liquid in a manner such that some of the liquid remains only in the vicinity of the surface of the protrusions adjacent to the shadowed region. a thin barrier layer may, for example, be applied to the surface prior to application of the liquid to protect against damage. a barrier layer applied to protect against damage may, for example, include at least one of an inorganic material, an organic material, or a metallic material. an inorganic material may,

▼ Pipeline inspection

Collection: patent_rag_chunks

Index path: /home/patzer_adi/Documents/python/Python_Workspace/Patent_Intelligence_Project/data/

Embedding model: sentence-transformers/all-MiniLM-L6-v2

chroma_db

Reranker model: cross-encoder/ms-marco-MiniLM-L-6-v2

Stage timings

▼

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"embedding" : 0.14303213599487208

"vector_search" : 0.5928880949941231

"reranking" : 2.920746545001748

}

Top retrieved chunks

embedding_rank	patent_number	chunk_index	embedding_similarity	reranker_rank	reranker_score
1	14905300	20	0.2492	1	7.3512
2	14903677	39	0.2488	2	5.773
4	14903677	6	0.1961	3	5.1977
22	14991299	33	0.0559	4	4.8039
8	14903677	8	0.1535	5	4.6136
3	14903677	0	0.2266	6	4.591
43	14991299	31	0.0024	7	4.5661
44	14906540	45	0.0015	8	4.491
32	15002057	13	0.0273	9	4.2834
95	14906379	42	-0.0641	10	3.6857